AGRIVOLTAICS AS MULTIFUNCTIONAL ENERGY INFRASTRUCTURE: PATHWAYS, BARRIERS, AND DESIGN PRINCIPLES FOR SCALABLE DUAL-USE LAND SYSTEMS

Extended Abstract

Teja Hanumolu [1], Vijay Paidi [2,*] , Ravi Vatrapu [3], Abid Hussain [4]

1 Teja Hanumolu; tejahanumolu123@gmail.com
2. Vijay Paidi; vpp@du.se *
3 Ravi Vatrapu; vatrapu@torontomu.ca
4 Abid Hussain; ah.digi@cbs.dk
* Correspondence: vpp@du.se

Extended Abstract

The transition toward low-carbon energy systems necessitates the rapid and large-scale deployment of renewable energy technologies, with solar photovoltaics (PV) playing a central role due to declining costs, technological maturity, and global availability [1]. At the same time, agriculture faces increasing pressures from climate change, population growth, and land degradation, leading to intensified competition for land resources. These overlapping challenges have raised concerns regarding land-use conflicts, food security, and rural livelihoods, particularly in regions with limited arable land.

Agrivoltaics (AV), also referred to as agrophotovoltaics, offers a potential solution by enabling the co-location of photovoltaic energy generation and agricultural production on the same land [2,3]. This dual-use approach departs from conventional land-use paradigms that treat energy and agriculture as mutually exclusive. Instead, agrivoltaics introduces a multifunctional framework that seeks to optimize combined land productivity. Over the past decade, increasing academic and industrial interest has generated a growing body of research demonstrating that AV systems can deliver a substantial share of the energy output associated with conventional PV installations while maintaining viable agricultural production [4,5].

In addition to supporting dual outputs, agrivoltaic systems can modify local microclimatic conditions in ways that benefit agricultural performance. Partial shading from PV modules has been shown to reduce soil temperatures, lower evapotranspiration rates, and improve water-use efficiency, particularly in hot and dry environments [6–8]. In certain contexts, these effects can stabilize or even enhance crop yields relative to open-field conditions. Despite these promising outcomes, agrivoltaics remains largely confined to pilot projects and early-stage deployments, concentrated in a limited number of countries [4]. Existing research often focuses on agronomic performance or technological design in isolation, with limited integration of energy systems, economic scalability, and governance considerations.

This extended abstract presents a concise synthesis of agrivoltaics research across four interrelated domains: system configurations and energy performance; land-use efficiency and agricultural–energy interactions; techno-economic performance and scalability; and policy, legal, and social constraints. This integrated perspective reframes agrivoltaics as

multifunctional energy infrastructure rather than a niche agricultural innovation.

Keywords: Agrivoltaics; dual-use land use; photovoltaic systems; land-use efficiency; energy food nexus.

1 Agrivoltaic System Configurations and Energy Performance

Agrivoltaic system design fundamentally shapes both energy generation and agricultural output. Unlike conventional PV installations, AV systems must balance solar capture with agricultural accessibility, leading to a range of configurations that reflect trade-offs among energy density, structural complexity, and crop compatibility.

Elevated agrivoltaic systems mount PV modules several meters above the ground, allowing conventional agricultural operations to continue beneath the array. These systems are often integrated with solar tracking technologies to optimize irradiance capture throughout the day and have been shown to achieve approximately 85–95% of the energy yield of conventional PV installations despite reduced module density [9,10]. In addition to supporting mechanized agriculture and high-value crops, elevated systems can moderate microclimatic conditions by reducing solar exposure, thereby lowering evapotranspiration and mitigating heat stress, particularly in arid regions [11]. However, these advantages come with increased structural requirements, higher capital costs, and greater engineering complexity, while uneven shading patterns and exposure to wind loads may further affect system performance and crop outcomes [12–15].

Spaced or row-based systems provide an alternative configuration by increasing the distance between PV rows rather than significantly elevating the panels. This approach allows greater sunlight penetration while maintaining relatively conventional mounting heights, making it easier to integrate into existing agricultural systems [16]. However, the reduction in panel density leads to lower overall energy output, and system performance becomes highly sensitive to design parameters such as spacing, tilt, and orientation, which directly influence both energy yield and crop exposure [15,17].

Vertical and bifacial configurations represent another approach, emphasizing spatial efficiency and operational compatibility. Typically oriented east–west, vertical panels create alternating bands of sunlight and shade, enabling easier movement of agricultural machinery and reducing interference with farming activities [18]. Although energy yield is generally lower than that of optimally tilted systems, bifacial modules can capture reflected radiation from the ground, partially compensating for reduced direct irradiance [16]. Across all configurations, performance remains highly dependent on site-specific factors such as climate, crop type, and local environmental conditions.

Taken together, these configuration-specific trade-offs illustrate that agrivoltaic system design is not a one-size-fits-all solution but instead requires context-sensitive optimization. Importantly, these trade-offs directly influence how effectively land can be utilized, highlighting the need to evaluate agrivoltaic systems not only in terms of energy performance but also in terms of integrated land-use efficiency.

2 Land Use Efficiency and Agricultural Energy Interactions

A defining advantage of agrivoltaics lies in its ability to enhance land-use efficiency by enabling the simultaneous production of food and energy on the same land area. This represents a departure from conventional land-use approaches, where agriculture and energy generation are treated as competing uses of land [2,19]. By integrating both functions, agrivoltaics reframes land productivity in terms of combined outputs rather than single-use optimization. In this context, evaluating how efficiently land is utilized becomes central to assessing agrivoltaic performance. The most widely used metric to evaluate this efficiency is the land equivalent ratio (LER), which compares the combined productivity of an AV system to that of separate monoculture systems. Values greater than one indicate improved land-use efficiency, and empirical studies frequently report LER values ranging from approximately 1.1 to 1.7 depending on system configuration, crop type, and environmental conditions [6,9,21]. However, despite its widespread use, LER is subject to methodological inconsistencies, including variations in baseline assumptions and land accounting methods, which complicate cross-study comparisons and limit its application in broader energy system planning [12,22,23].

A key mechanism underlying these outcomes is microclimatic modification. PV modules alter radiation, temperature, wind speed, and humidity at the soil surface, leading to reduced evapotranspiration and improved soil moisture retention [6,24]. These changes are particularly beneficial in hot and dry climates, where reduced thermal stress can help stabilize crop yields under increasingly variable climatic conditions [25].

Crop responses to agrivoltaic environments, however, vary significantly depending on species and growth stage. Shade-tolerant crops often benefit from reduced thermal stress and improved moisture conditions, whereas light-demanding crops may experience yield reductions if shading is not carefully managed [26,27]. In addition to yield effects, agrivoltaics can influence crop quality, phenology, and stress resilience, with outcomes depending on both crop characteristics and local environmental conditions [28,29].

These dynamics highlight the inherent trade-offs between energy density and agricultural exposure. Increasing PV coverage enhances electricity generation but reduces light availability for crops, while wider spacing improves agricultural conditions at the expense of energy output. As a result, optimal agrivoltaic design focuses on maximizing combined system productivity rather than prioritizing individual outputs in isolation.

3 Techno-Economic Performance and Scalability

The scalability of agrivoltaics depends not only on technical feasibility but also on its economic performance relative to conventional PV systems. A consistent finding across the literature is that AV systems incur higher capital expenditure due to increased structural complexity, greater material requirements, and the need for customized designs that maintain agricultural accessibility [9,30]. These cost premiums, which range from approximately 5% to over 30% depending on system configuration and regional conditions, vary significantly across elevated, spaced, and vertical designs, each of which introduces distinct trade-offs between cost, energy density, and agricultural compatibility [17,31]. At the same time, the

absence of standardized designs and economies of scale further contributes to cost variability, complicating direct comparisons with conventional PV systems [32].

The levelized cost of electricity (LCOE) remains a commonly used metric for evaluating PV systems; however, it is insufficient for agrivoltaics because it does not account for agricultural outputs [33]. As a result, AV systems may appear less competitive when assessed using electricity-only metrics, despite their ability to generate multiple forms of value. To address this limitation, researchers increasingly advocate for multi-output economic frameworks that integrate both energy and agricultural revenues [34–36]. When these additional value streams are considered, agrivoltaics can achieve competitive or even superior economic performance, particularly in regions with high land values or strong agricultural productivity [37,38]. In this context, agrivoltaics not only enhances land productivity but also provides a more diversified economic model.

This diversification is further reflected in risk management. Agrivoltaic systems combine relatively stable electricity revenues with more variable agricultural income, allowing each to partially offset uncertainties in the other [39]. However, despite these advantages, AV systems often exhibit longer payback periods, typically ranging from 7 to 12 years, compared to 5 to 8 years for conventional PV systems [30].

Scalability is also constrained by the site-specific nature of agrivoltaic design and the limited integration of AV systems into existing energy system models. Because agrivoltaics may reduce installed capacity density compared to conventional solar farms, it is often undervalued in planning frameworks that prioritize electricity output alone [40]. This highlights the need for more comprehensive evaluation approaches that capture the full range of benefits associated with dual-use systems.

4 Policy, Legal, and Social Constraints

Non-technical factors represent major barriers to the large-scale deployment of agrivoltaics. A key challenge is the fragmentation between energy and agricultural policy domains, where land is typically classified as either agricultural or industrial, leaving limited regulatory space for dual-use systems [41]. This classification creates legal ambiguity and complicates permitting processes. Regulatory authorities often require evidence that agricultural productivity will not be compromised; however, such assessments are inherently uncertain [42]. These challenges disproportionately affect small- and medium-scale farmers, who may lack the resources to navigate complex regulatory environments [43].

In some regions, zoning laws prohibit PV installations on agricultural land, while agricultural subsidy frameworks may exclude land used for energy production. These policy misalignments create economic disincentives for agrivoltaics [44].

Countries that explicitly recognize agrivoltaics as a distinct land-use category demonstrate more favorable conditions for deployment. For example, regulatory frameworks in Germany and Japan allow continued agricultural use under AV systems, reducing uncertainty and facilitating investment [45,46].

Social acceptance also plays a critical role. Concerns related to visual impact, land control, and the perceived prioritization of energy over agriculture can generate resistance [35]. However, projects that involve farmers and local stakeholders in planning and ownership tend to achieve higher levels of acceptance and long-term stability [47].

Equity considerations further influence access to agrivoltaics. Participation is often shaped by land ownership, access to capital, and technical expertise, potentially reinforcing existing inequalities in rural communities [48–50].

5 Conclusion and Future Directions

Overall, agrivoltaics offers a promising pathway for addressing land-use conflicts in renewable energy deployment by enabling the co-production of electricity and agricultural outputs on the same land. Evidence suggests that AV systems can achieve between 75% and 95% of conventional PV energy output while maintaining viable agricultural production [9,10]. Their primary advantage lies in improved land-use efficiency and enhanced resilience through microclimatic modification. However, large-scale adoption remains constrained by higher capital costs, a lack of standardized evaluation frameworks, and limited integration into energy system planning. More critically, fragmented policy environments, regulatory uncertainty, and social acceptance challenges represent significant barriers to deployment.

Future research should prioritize the development of standardized performance metrics, long-term empirical studies, integration into energy system models, and systematic analysis of governance frameworks. Greater attention to equity and stakeholder participation will also be essential to ensure inclusive and sustainable implementation. Agrivoltaics should therefore be understood not as a niche agricultural innovation but as a multifunctional energy infrastructure capable of contributing to decarbonization while preserving agricultural productivity.

References

1. Ugochukwu, A.A.; Ahmad, F.; Khalid, M.; Ali, M.A.; Mamoon, I.; Udensi, S.C.; Ogbonna, U.; Iqbal, A.; Ali, R.; Usman, M.; et al. **Recent Enhancement in Photovoltaic Cell Efficiency Performance, Stability, and Cost Reduction: A Review.** *Solar Energy* 2025, 300, 113853. https://doi.org/10.1016/j.solener.2025.113853
2. Goetzberger, A.; Zastrow, A. **On the Coexistence of Solar-Energy Conversion and Plant Cultivation.** *International Journal of Solar Energy* 1982, 1, 55–69. https://doi.org/10.1080/01425918208909875
3. Dinesh, H.; Pearce, J.M. **The Potential of Agrivoltaic Systems.** *Renewable and Sustainable Energy Reviews* 2016, 54, 299–308. https://doi.org/10.1016/j.rser.2015.10.024
4. Chopdar, R.K.; Sengar, N.; Giri, N.C.; Halliday, D. **Comprehensive Review on Agrivoltaics with Technical, Environmental and Societal Insights.** *Renewable and Sustainable Energy Reviews* 2024, 197, 114416. https://doi.org/10.1016/j.rser.2024.114416
5. Cho, J.; Park, S.M.; Park, A.R.; Lee, O.C.; Nam, G.; Ra, I.-H. **Application of Photovoltaic Systems for Agriculture: A Study on the Relationship between Power Generation and Farming for the Improvement of Photovoltaic Applications in Agriculture.** *Energies* 2020, 13, 4815. https://doi.org/10.3390/en13184815

6. Marrou, H.; Guilioni, L.; Dufour, L.; Dupraz, C.; Wery, J. **Microclimate under Agrivoltaic Systems: Is Crop Growth Rate Affected in the Partial Shade of Solar Panels?** *Agricultural and Forest Meteorology* 2013, 177, 117–132. https://doi.org/10.1016/j.agrformet.2013.04.012
7. Ali Abaker Omer, A.; Li, M.; Zhang, F.; Hassaan, M.M.E.; El Kolaly, W.; Zhang, X.; Lan, H.; Liu, J.; Liu, W. **Impacts of Agrivoltaic Systems on Microclimate, Water Use Efficiency, and Crop Yield: A Systematic Review.** *Renewable and Sustainable Energy Reviews* 2025, 221, 115930. https://doi.org/10.1016/j.rser.2025.115930
8. Randle-Boggis, R.J.; Lara, E.; Onyango, J.; Temu, E.J.; Hartley, S.E. **Agrivoltaics in East Africa: Opportunities and Challenges.** *AIP Conference Proceedings* 2021, 2361, 090001. https://doi.org/10.1063/5.0055470
9. Trommsdorff, M.; Kang, J.; Reise, C.; Schindele, S.; Bopp, G.; Ehmann, A.; Weselek, A.; Högy, P.; Obergfell, T. **Combining Food and Energy Production: Design of an Agrivoltaic System Applied in Arable and Vegetable Farming in Germany.** *Renewable and Sustainable Energy Reviews* 2021, 140, 110694. https://doi.org/10.1016/j.rser.2020.110694
10. Trommsdorff, M.; Dhal, I.S.; Özdemir, Ö.E.; Ketzer, D.; Weinberger, N.; Rösch, C. **Agrivoltaics: Solar Power Generation and Food Production.** In *Solar Energy Advancements in Agriculture and Food Production Systems*; Elsevier: Amsterdam, The Netherlands, 2022; pp. 159–210.
11. Teng, J.W.C.; Soh, C.B.; Devihosur, S.C.; Tay, R.H.S.; Jusuf, S.K. **Effects of Agrivoltaic Systems on the Surrounding Rooftop Microclimate.** *Sustainability* 2022, 14, 7089. https://doi.org/10.3390/su14127089
12. Willockx, B.; Cappelle, J. **A Comparison Study of the Performance of Vertical vs Single Axis Tracking Bifacial Agrivoltaic Systems in Belgium.** In *Proceedings of the 2022 IEEE 49th Photovoltaic Specialists Conference (PVSC)*; IEEE: Philadelphia, PA, USA, 2022; pp. 1162–1162.
13. Krexner, T.; Bauer, A.; Gronauer, A.; Mikovits, C.; Schmidt, J.; Kral, I. **Environmental Life Cycle Assessment of a Stilted and Vertical Bifacial Crop-Based Agrivoltaic Multi Land-Use System and Comparison with Mono Land-Use.** *Renewable and Sustainable Energy Reviews* 2024, 196, 114321. https://doi.org/10.1016/j.rser.2024.114321
14. Tina, G.M.; Osama, A.; Cazzaniga, R.; Cicu, M.; Hancock, J.; Howlin, E.; Rosa-Clot, M.; Rosa-Clot, P. **PVSails: Harnessing Innovation with Vertical Bifacial PV Modules in Floating Photovoltaic Systems.** *Progress in Photovoltaics: Research and Applications* 2024, 32, 872–888. https://doi.org/10.1002/pip.3841
15. Riaz, M.H.; Imran, H.; Younas, R.; Alam, M.A.; Butt, N.Z. **Module Technology for Agrivoltaics: Vertical Bifacial Versus Tilted Monofacial Farms.** *IEEE Journal of Photovoltaics* 2021, 11, 469–477. https://doi.org/10.1109/JPHOTOV.2020.3048225
16. Carrausse, R.; Arnauld de Sartre, X. **Does Agrivoltaism Reconcile Energy and Agriculture? Lessons from a French Case Study.** *Energy, Sustainability and Society* 2023, 13, 8. https://doi.org/10.1186/s13705-023-00387-3
17. Campana, P.E.; Stridh, B.; Amaducci, S.; Colauzzi, M. **Optimisation of Vertically Mounted Agrivoltaic Systems.***Journal of Cleaner Production* 2021, 325, 129091. https://doi.org/10.1016/j.jclepro.2021.129091
18. Riaz, M.H.; Imran, H.; Alam, H.; Alam, M.A.; Butt, N.Z. **Crop-Specific Optimization of Bifacial PV Arrays for Agrivoltaic Food-Energy Production: The Light-Productivity-Factor Approach.** *IEEE Journal of Photovoltaics*2022, 12, 572–580. https://doi.org/10.1109/JPHOTOV.2021.3136158
19. Dupraz, C.; Marrou, H.; Talbot, G.; Dufour, L.; Nogier, A.; Ferard, Y. **Combining Solar Photovoltaic Panels and Food Crops for Optimising Land Use: Towards New Agrivoltaic Schemes.** *Renewable Energy* 2011, 36, 2725–2732. https://doi.org/10.1016/j.renene.2011.03.005
20. Garrod, A.; Hussain, S.N.; Ghosh, A. **The Technical and Economic Potential for Crop-Based Agrivoltaics in the United Kingdom.** *Solar Energy* 2024, 277, 112744. https://doi.org/10.1016/j.solener.2024.112744

21. Birchall, M.; Hussain, S.N.; Ghosh, A. **Review of Agrivoltaic Demonstration Site Studies with Comparable Configurations for a UK-Based Application.** *Solar Compass* 2025, 15, 100135. https://doi.org/10.1016/j.solcom.2025.100135
22. Campana, P.E.; Stridh, B.; Hörndahl, T.; Svensson, S.-E.; Zainali, S.; Lu, S.M.; Zidane, T.E.K.; De Luca, P.; Amaducci, S.; Colauzzi, M. **Experimental Results, Integrated Model Validation, and Economic Aspects of Agrivoltaic Systems at Northern Latitudes.** *Journal of Cleaner Production* 2024, 437, 140235. https://doi.org/10.1016/j.jclepro.2023.140235
23. Trommsdorff, M.; Hopf, M.; Hörnle, O.; Berwind, M.; Schindele, S.; Wydra, K. **Can Synergies in Agriculture through Integration of Solar Energy Reduce Costs?** *Applied Energy* 2023, 350, 121619. https://doi.org/10.1016/j.apenergy.2023.121619
24. Adeh, E.H.; Selker, J.S.; Higgins, C.W. **Remarkable Agrivoltaic Influence on Soil Moisture, Micrometeorology and Water-Use Efficiency.** *PLOS ONE* 2018, 13, e0203256. https://doi.org/10.1371/journal.pone.0203256
25. Barron-Gafford, G.A.; Pavao-Zuckerman, M.A.; Minor, R.L.; Sutter, L.F.; Barnett-Moreno, I.; Blackett, D.T.; Thompson, M.; Dimond, K.; Gerlak, A.K.; Nabhan, G.P.; et al. **Agrivoltaics Provide Mutual Benefits across the Food–Energy–Water Nexus in Drylands.** *Nature Sustainability* 2019, 2, 848–855. https://doi.org/10.1038/s41893-019-0364-5
26. Widmer, J.; Christ, B.; Grenz, J.; Norgrove, L. **Agrivoltaics: A Promising Tool for Electricity and Food Production.***Renewable and Sustainable Energy Reviews* 2024, 192, 114277.
27. Weselek, A.; Ehmann, A.; Zikeli, S.; Lewandowski, I.; Schindele, S.; Högy, P. **Agrophotovoltaic Systems: Applications, Challenges, and Opportunities.** *Agronomy for Sustainable Development* 2019, 39, 35.
28. Schulz, V.S.; Munz, S.; Stolzenburg, K.; Hartung, J.; Weisenburger, S.; Graeff-Hönninger, S. **Impact of Different Shading Levels on Growth, Yield and Quality of Potato.** *Agronomy* 2019, 9, 330.
29. Qin, F.; Shen, Y.; Li, Z.; Qu, H.; Feng, J.; Kong, L.; Teri, G.; Luan, H.; Cao, Z. **Shade Effects on Phenology and Growth.** *Frontiers in Plant Science* 2022, 13, 835380.
30. Schindele, S.; Trommsdorff, M.; Schlaak, A.; Obergfell, T.; Bopp, G.; Reise, C.; Braun, C.; Weselek, A.; Bauerle, A.; Högy, P.; et al. **Techno-Economic Analysis of Agrivoltaics.** *Applied Energy* 2020, 265, 114737.
31. Zainali, S.; Qadir, O.; Parlak, S.C.; Lu, S.M.; Avelin, A.; Stridh, B.; Campana, P.E. **CFD Modelling of Microclimate for Vertical AV Systems.** *Energy Nexus* 2023, 9, 100173.
32. Campana, P.E.; Stridh, B.; Zainali, S.; et al. **Evaluation of Agrivoltaic Systems in Sweden.** 2023.
33. Willockx, B.; Lavaert, C.; Cappelle, J. **Geospatial Assessment of Elevated Agrivoltaics in Europe.** *Energy Reports*2022, 8, 8736–8751.
34. Neupane Bhandari, S.; Schlüter, S.; Kuckshinrichs, W.; Schlör, H.; Adamou, R.; Bhandari, R. **Economic Feasibility of Agrivoltaic Systems.** *Agronomy* 2021, 11, 1906.
35. Tomasi, S.; Pellegrini, C.; Gaspari, F.; et al. **Agrivoltaics and Rural Acceptance.** *Energy Research & Social Science*2025, 129, 104369.
36. Nannetti, M. **Techno-Economic Assessment of Agrivoltaic Case Studies.** Master's Thesis, Politecnico di Milano, 2023.
37. Cuppari, R.I.; Fernandez-Bou, A.S.; Characklis, G.W.; et al. **Drivers of Agrivoltaic Perception.** *Environmental Research: Food Systems* 2024, 1, 021003.
38. Farja, Y.; Maciejczak, M. **Economic Implications of Land Conversion.** *Energies* 2021, 14, 6063.
39. Cuppari, R.I.; Higgins, C.W.; Characklis, G.W. **Agrivoltaics and Weather Risk Diversification.** *Applied Energy* 2021, 291, 116809.
40. Ali Khan Niazi, K.; Victoria, M. **Comparative Analysis of PV Configurations.** *Progress in Photovoltaics* 2023, 31, 1101–1113.
41. Vollprecht, J.; Battoue, L.; Bois-Minot, A.; et al. **Legal Frameworks in Agrivoltaics.** IET, 2024.

42. Taylor, M.; McDonnell, N.; Davies, P.; Trück, S. **Scaling Agrivoltaics: Planning and Legal Pathways.** *Sustainability Science* 2025, 20, 1499–1517.
43. Crowther, A.; Foulds, C.; Robison, R.; et al. **Energy Policy Governance Recommendations.** Springer, 2024.
44. Trommsdorff, M.; Campana, P.E.; Macknick, J.; et al. **Dual Land Use for Agriculture and Solar Power.** IEA PVPS, 2025.
45. Jamil, U.; McDonnell, N.; Davies, P.; Taylor, M.; Pearce, J.M. **Just Agrivoltaic Transitions.** *Solar Energy* 2026, 305, 114195.
46. Doedt, C.; Tajima, M.; Iida, T. **Agrivoltaics in Japan: Legal Framework Analysis.** 2022.
47. Walston, L.J.; Barley, T.; Bhandari, I.; et al. **Sustainability Opportunities of Agrivoltaics.** *Frontiers in Sustainable Food Systems* 2022.
48. Pandey, G.; Lyden, S.; Franklin, E.; et al. **Systematic Review of Agrivoltaics.** *Sustainable Production and Consumption* 2025, 56, 13–36.
49. Hayibo, K.S.; Jamil, U.; Pearce, J.M. **Co-Benefits of Dual-Use Solar Systems.** *Renewable and Sustainable Energy Reviews* 2025, 222, 116021.
50. Mehta, K.; Betz, F. **Geospatial Suitability for Agrivoltaics in India.** *Energy for Sustainable Development* 2026, 91, 101915.